

Newnal

Intellectual Property Rights

*List of Technical Patents, Trademarks, Design Patents,
and Technical Whitepapers*

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1. Patent Application and Registration Status of Blockchain Labs, Newnal Inc.

Patent Category	Patent Office	Patent Applications	Patent Registrations
Technical Patents	USPTO (United States)	22	6
	WIPO / PCT (International)	5	5
	EPO (Europe)	1	1
	KIPO (Korea)	30	19
	(Total)	56	31
Trademarks	USPTO (United States)	3	2
	EPO (Europe)	1	0
	KIPO (Korea)	9	7
	WIPO (International)	1	1
	Other Countries	4	4
	(Total)	18	14
Design Patents	USPTO (United States)	3	2
	KIPO (Korea)	7	7
	CNIPA (China)	2	2
	(Total)	11	10
Total		85	55

2. Technical Patents

2.1 Core Patents

A) Core Blockchain Technology (InfraBlockchain)

BLOCKCHAIN SYSTEM TO WHICH PROOF-OF-TRANSACTION CONSENSUS ALGORITHM IS APPLIED, AND METHOD THEREFOR	
World's first developed Proof-of-Transaction (PoT) blockchain consensus algorithm technology that enables the establishment of a nationwide scalable stablecoin-based blockchain network without issuing virtual cryptocurrency	
[USPTO] US-11636450-B2	
[WIPO/PCT] WO/2019/182202	
[KIPO] 10-225081	

MULTI BLOCKCHAIN NETWORK SYSTEM WITHOUT NATIVE CRYPTOCURRENCY BASED ON AGGREGATED PROOF-OF-TRANSACTION CONSENSUS
Aggregated Proof-of-Transaction (PoT) multi-Chain consensus algorithm technology patent that allows a large number of multi-blockchain networks to be operated without issuing virtual cryptocurrency. A technology that enables the establishment of a super large-scale multi-blockchain network consisting of multiple blockchain networks operating in parallel

[USPTO] US-20240330915-A1	
[KIPO] 10-2023-0042651	

THE TRADING SYSTEM AND THE METHOD BASED ON A BLOCKCHAIN	
Large-scale high frequency trading system and method built on blockchain for secure, verifiable on-chain/off-chain hybrid transactions	
[WIPO/PCT] WO/2019/035573	
[KIPO] 10-2309819	

B) Blockchain-based Secure Communication & Messaging

ELECTRONIC DEVICE AND METHOD PROVIDING SECRET MESSENGER FUNCTION THROUGH END TO END ENCRYPTION BASED ON BLOCKCHAIN DID TECHNOLOGY	
Electronic device and method delivering a secret messenger with end-to-end encryption anchored in blockchain DID technology for verifiable, private communication	
[USPTO] US-19/104055	
[WIPO/PCT] WO/2024/048838	
[KIPO] 10-2773087	

METHOD AND DEVICE FOR BUILDING DECENTRALIZED SECURE COMMUNICATION INFRASTRUCTURE NETWORK BASED ON BLOCKCHAIN AND DID TECHNOLOGY	
Technology patent for the construction of a scalable blockchain-based decentralized federated security communication infrastructure node network using a DID-based phone number generation mechanism	
[USPTO] US-19/059751	
[KIPO] 10-2025-0017836	

C) Personal Data Ownership & Web3 Data Infrastructure

BLOCKCHAIN-BASED WEB DATA OWNERSHIP AND DATA ACCESS MANAGEMENT METHOD AND SYSTEM FOR TRACKING AI DATA COLLECTION	
Blockchain system that establishes verifiable ownership of web data and manages access, letting individuals track and control how AI collects and uses their data	
[USPTO] US-20250055712-A1	
[EPO] EP-23195338.1	
[KIPO] 10-2023-0103842	

**DEVICE FOR PROVIDING BLOCKCHAIN DID-BASED MULTI CLOUD SERVICE
AND ITS OPERATING METHOD**

Technology patent for building a blockchain-based system that provides blockchain and DID-based multi-cloud services to distribute and store personal data in encrypted form in multiple cloud environments with the decentralized identity (DID) of individual users and manage personal data safely

[USPTO] US-20240414014-A1



[KIPO] 10-2023-0073929



**A DEVICE AND METHOD FOR OWNING AND TRADING VERIFIABLE PERSONAL
DATA THROUGH BLOCKCHAIN BASED PERSONAL WEB NODES**

Technology patent for building a system that allows individuals to store verifiable personal data and have data ownership through blockchain-based private web nodes, granting data sovereignty to users and enabling them to trade personal data securely

[USPTO] US-19/038309



[KIPO] 10-2024-0012155



D) Personalized AI Model & AI Agent

PERSONALIZED AI ACTION GRAPH GENERATION MODEL BASED AI INFERENCE DEVICE	
An AI inference device that builds a personalized action-graph model from an individual's data to predict and act on user-specific intent and behavior	
[USPTO] US-18/779899	
[KIPO] 10-2024-0090806	

DECENTRALIZED AI AGENT NETWORK SYSTEM BUILT UPON BLOCKCHAIN-BASED WEB3 AI-ID AND WEB3 AI-CONTRACT FOR VERIFIABLE CONFIDENTIAL AI AGENT COMPUTING	
A decentralized AI agent network built on blockchain Web3 AI-ID and AI-Contract, enabling verifiable and confidential computing between autonomous AI agents	
[USPTO] US-19/303959 US-63/764594	
[KIPO] 10-2025-0107438	

BLOCKCHAIN-BASED AI AGENT SERVICE MARKETPLACE SYSTEM INVOLVING PERSONAL AI AGENTS AND SERVICE AI AGENTS UTILIZING PERSONAL DATA

Technology patent for building a blockchain-based marketplace platform where personal AI agents and service AI agents trade personal data through open markets and exchange personalized services based on personal data

[USPTO] US-19/702994
US-63/865619



[KIPO] 10-2026-0079294



A DEVICE AND METHOD FOR EXTRACTING AND STRUCTURING VERIFIABLE PERSONAL DATA FROM USER INPUT AND OUTPUT DATA CAPTURED ON A DEVICE BASED ON A MULTI-MODAL AND LANGUAGE AI MODEL

A device and method that uses multi-modal and language AI models to extract and structure verifiable personal data from a user's on-device inputs and outputs.

[USPTO] US-19/041098



[KIPO] 10-2024-0014525



E) Personal AI Operating System & Devices

MOBILE DEVICE FOR OPERATING PERSONALIZED ON-DEVICE AI AGENT BASED ON PERSONAL DATA	
A mobile device that runs a personalized AI agent locally on the user's personal data, keeping inference and personalization on-device for privacy and data control	
[USPTO] US-19/059568	
[KIPO] 10-2025-0019132	

USER INTERFACE FOR RUNNING A PERSONALIZED AI-DEDICATED OPERATING SYSTEM ON VARIOUS TYPES OF USER DEVICES	
A user interface that enables a personalized AI-dedicated operating system to run consistently across diverse types of user devices	
[USPTO] US-19/277871	
[KIPO] 10-2025-0098140	

BLOCKCHAIN-BASED AI COMPUTING MOBILE OPERATING SYSTEM FOR SECURE DECENTRALIZED PERSONAL DATA AND PERSONALIZED AI AGENT SERVICES	
A blockchain-based mobile OS for secure, decentralized handling of personal data and the delivery of personalized AI agent services on the device	
[USPTO] US-63/884059	

F) Decentralized Identity (DID) & Verifiable Credentials

<i>DID AUTHENTICATION METHOD USING SMART CARD AND SMART CARD DEVICE</i>	
A DID authentication method and smart card device that enable decentralized identity verification through a physical secure card	
[USPTO] US-18/709142	
[WIPO/PCT] WO/2023/085802	
<i>METHOD AND SYSTEM FOR PROVIDING CERTIFICATION OF VACCINE INOCULATION AND POST-INOCULATION MANAGEMENT</i>	
A method and system for issuing and verifying vaccine inoculation certificates and managing post-inoculation records using blockchain-based verifiable credentials.	
[USPTO] US-20240185994-A1	
[WIPO/PCT] WO/2022/196851	
[KIPO] 10-2488139	
<i>IDENTITY MANAGEMENT SYSTEMS AND METHODS USING DID BASED ON BLOCKCHAIN TECHNOLOGY</i>	
Identity management systems and methods that use decentralized identifiers (DID) on blockchain to give users self-sovereign control over their identity.	
[KIPO] 10-2022-0100263	

2.2 Periphery Patents

[USPTO] US-63/763955

A METHOD AND SYSTEM FOR COLLECTING AND INTEGRATING PAST, PRESENT AND FUTURE PERSONAL DATA TO IMPLEMENT PERSONALIZED AI MODEL AND AI AGENT ENGINE



[KIPO] 10-2024-0014525

A DEVICE AND METHOD FOR EXTRACTING AND STRUCTURING VERIFIABLE PERSONAL DATA FROM USER INPUT AND OUTPUT DATA CAPTURED ON A DEVICE BASED ON A MULTI-MODAL AND LANGUAGE AI MODEL



[USPTO] US-19/303959

METHOD AND DEVICE FOR CONFIGURING AND OPERATING SELF-EVOLVING PERSONAL AI AGENT BASED ON PERSONAL DATA



[KIPO] 10-2026-0012833

METHOD AND DEVICE FOR CONFIGURING AND OPERATING SELF-EVOLVING PERSONAL AI AGENT BASED ON PERSONAL DATA



[USPTO] US-19 (application in progress)

BLOCKCHAIN-BASED HOME NODE DEVICE AND SERVER-BASED OPERATING SYSTEM FOR PROVIDING PERSONAL DATA SERVICES AND PERSONAL AI AGENT SERVICES



[KIPO] 10-2026-0013613

BLOCKCHAIN-BASED HOME NODE DEVICE AND SERVER-BASED OPERATING SYSTEM FOR PROVIDING PERSONAL DATA SERVICES AND PERSONAL AI AGENT SERVICES



[USPTO] US-63/763302

METHOD AND SYSTEM FOR ADAPTIVE FACIAL REGION BLENDING IN VIDEO SYNTHESIS



[KIPO] 10-2249864

SERVER AND METHOD FOR CREDIT TRANSACTION USING BLOCKCHAIN NETWORK



[KIPO] 10-2023-0165942

HIGH-PERFORMANCE CONSENSUS ALGORITHM SIMULATOR



[KIPO] 10-2700653

THE SYSTEM AND THE METHOD FOR MANAGING DISTRIBUTION HISTORY USING NON-FUNGIBLE TOKEN



[KIPO] 10-2022-0130057

SYSTEM AND METHOD FOR TRADING BASED ON BLOCKCHAIN TECHNOLOGIES



[KIPO] 10-2613633

AI MODEL TRAINING APPARATUS AND METHOD FOR READING TEST RESULTS OF A DIAGNOSTIC KIT



[KIPO] 10-2478963

A SYSTEM AND METHOD FOR ISSUING AND VERIFYING DIGITAL VACCINATION CERTIFICATES



[KIPO] 10-2490640

ELECTRONIC DEVICE AND METHOD FOR PROVING PRIVATE INFORMATION BETWEEN INDIVIDUALS BASED ON BLOCKCHAIN TECHNOLOGY



[KIPO] 10-2591773

METHOD AND SYSTEM FOR CERTIFICATING COVID-19 ANTIBODY



[KIPO] 10-2022-0100264

AI-BASED IDENTITY RECOGNITION METHOD AND ELECTRONIC DEVICE THEREOF



[KIPO] 10-2022-0108933

BLOCKCHAIN DID-BASED REAL-TIME REAL ESTATE REGISTRATION MANAGEMENT SYSTEM



[KIPO] 10-2023-0051359

VEHICLE REMOTE CONTROL AND VEHICLE INFORMATION MANAGEMENT METHOD USING VEHICLE-DID BASED ON BLOCKCHAIN NETWORKS AND THE VEHICLE DEVICE THEREOF



3. Trademarks

Title	Class	Application Number
Newnal	09, 35, 36, 38, 42	[US] US 98/180961 [EU] EP 18943637 [CH,GB] WIPO 1773137 [KR] 40-2023-0167191
Metacircle	9	[US] US 99/290317 [EU] EP 19219923 [KR] 40-2025-0125955
COOV	09,42	[KR] 40-2021-0065186 [KR] 40-2021-0065182
InfraBlockchain	42	[KR] 40-2023-0122739
BlockChat	09, 38, 42	[KR] 40-2022-0139282 [KR] 40-2022-0139284 [KR] 40-2022-0207516 [GB,FR,ES,DE,IT,NL,BR,AU,AT,TR,RU,PT,ID,IN,SA,PH] WIPO 1710775 [HK] 306125436 [AR] 4196441 [TW] 111090047
BLOCKCHAIN CHIP Powered by Blockchain Labs	9	[KR] 40-2023-0100452

4. Design Patents

[KIPO] 30-2024-0004156

Devices with a rollable display



[KIPO] 30-2024-0004155

Devices with a rollable display – unrolled



[KIPO] 30-2024-0004155

Devices with a rollable display – rolled



[KIPO] 30-2025-0015837

Mobile phone device (Newnal Phone V1)



[KIPO] 30-2025-0015838

Mobile phone device – AI jog button (Newnal Phone V1)



[USPTO] US-30/015605

Mobile phone device (Newnal Phone V1)



[CNIPA] 202530452354.9

Mobile phone device (Newnal Phone V1)



[CNIPA] 202530510789.4

Mobile phone device (Newnal Phone Light)



[KIPO] 30-2025-0029489

Mobile phone device (Newnal Phone Light)



[USPTO] US-30/018308

Mobile phone device (Newnal Phone Light)



[KIPO] 30-2025-28631

Newnal Metacircle UI design – Homescreen, Lockscreen



[USPTO] US-30/015673

Newnal Metacircle UI design – Homescreen, Lockscreen



5. Technical Whitepapers

Newnal Web3 ai Technical Whitepaper

'My own ai' (another-i) model that learns from personal data

Technical Document  

Newnal Web3 ai OS Technical Whitepaper

Blockchain-based 'Appless OS' architecture for personal AI agents having Web3 ai-ID and Web3 ai-Contract

Technical Document  

Newnal Web3 Telecom Technical Whitepaper

Next-generation decentralized and secure communication protocol including AI-involved calls

Technical Document  

Newnal ai Agent Place Technical Whitepaper

Marketplace for personal data and ai(another-i) agent service exchange

Technical Document  

Newnal InfraBlockchain Technical Whitepaper

Stablecoin-native public federated multi-blockchain network and verifiable web data infrastructure

Technical Document  

Universal Resource Auth (UR-Auth) Protocol and System Architecture Technical Whitepaper

Blockchain-based Web-scale Data Ownership Management Technology

Technical Document  

InfraBlockchain Technical Whitepaper

Stablecoin as system tx fee token, Transaction-as-a-Vote, Proof-of-Transaction(PoT) consensus algorithm

Technical Document  

ON/OFF-Chain Hybrid Token Exchange System Technical Whitepaper

Blockchain-based large-scale high frequency trading system technology

Technical Document  

InfraBlockchain-Substrate Multi-Blockchain Developer Documentation

Technical Document  

InfraBlockchain-Antelope Developer Documentation

Technical Document  

InfraDID Method Specification



InfraDID W3C DID Spec. Registry (did:infra)



InfraBlockchain Open-Source Technology

